Hierarchical Structure-from-Motion Scales Feedforward Reconstruction

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Abstract. 3D foundation models have opened a new era for scene reconstruction. By leveraging large-scale pretraining, they bypass traditional Structure-from-Motion (SfM) pipelines and achieve feed-forward, generalizable performance across diverse scenes. However, their scalability remains limited by GPU memory constraints when applied to larger scenes. Several efforts have been made to mitigate these issues, such as developing more efficient model architectures or sequentially feeding smaller portions of the scene into the model. Nevertheless, scalability and flexibility remain challenging—particularly when reconstructing from large-scale, unstructured image collections. Therefore, we introduce a global reconstruction framework that constructs a verified visibility graph using cross-image retrieval and then computes a Nested Dissection (ND) based ordering to create a hierarchical tree of smaller, overlapping subgraphs. Each subgraph is reconstructed independently using a 3D foundation model, and the resulting local reconstructions are merged through a hierarchical bundle adjustment that jointly optimizes camera poses and 3D structures via shared cameras and correspondences. We demonstrate the effectiveness of our method on large scenes with VGGT-Omega, achieving competitive and, in several cases, state-of-the-art reconstruction accuracy on standard benchmarks.

Keywords: Structure-from-Motion · Multi-View 3D reconstruction · 3D Foundation Models · Hierarchical Graph Partitioning · Robust Bundle Adjustment

1 Introduction

Reconstructing accurate 3D scenes from large-scale unstructured image collections is a fundamental problem in computer vision with broad applications in digital twins, urban modeling, robotics, self driving, etc. As image collections continue to grow from hundreds to tens or even hundreds of thousands of images, scalability becomes one of the primary challenges for existing 3D reconstruction systems. Classical Structure-from-Motion (SfM) pipelines achieve

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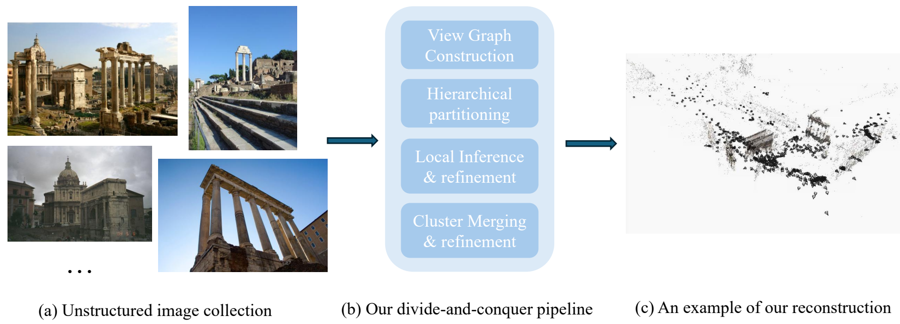

Fig. 1: Hierarchical Structure-from-Motion Scales Feedforward Reconstruction Given an unstructured image collection (a) with example images from [41], we construct a verified view graph and recursively partition it into smaller subgraphs. Each partition is reconstructed independently using a 3D foundation model and refined locally, before being hierarchically merged into a globally consistent reconstruction (b). Our divide-and-conquer framework enables scalable and parallel large-scale reconstruction. An example output is shown in (c).

high reconstruction accuracy through expensive multi-stage optimization methods, but these optimization-heavy pipelines become increasingly expensive as the scene grows, often requiring hours or even days to reconstruct large-scale datasets [2, 32, 33].

Recent trends in feedforward multi-view 3D reconstruction models have demonstrated a promising alternative. Models such as VGGT [35], MapAnything [19], π^3 [39], Depth Anything V3 [23], and VGGT-Ω [36] can directly infer camera poses, dense geometry, and in some cases multi-view tracks in a single feedforward pass. This dramatically reduces reconstruction time while exhibiting impressive robustness across challenging scenes (low multi-view overlap, repetitive structure, etc.). These works suggest that many traditional optimization based stages can instead be replaced by learned end-to-end models.

Despite this progress, optimization-based SfM is still the method of choice for large-scale unstructured reconstruction, due to multiple reasons. First, while feedforward models are fast, their geometric accuracy is still inferior to that of optimization-based methods (except in challenging cases with insufficient overlap). Second, existing transformer-based architectures are fundamentally limited by memory, making it difficult to process thousands of images jointly. Even if computational memory constraints are to be ignored, there is evidence that the reconstruction accuracy of feedforward models worsens with increasing sequence length [28]. Recent autoregressive approaches alleviate the quadratic memory complexity by processing images sequentially, but they are designed primarily for continuous video streams with small baselines, consistent appearance, and shared camera intrinsics. Consequently, they are not well suited for large, unordered Internet photo collections exhibiting wide viewpoint changes, significant appearance variation, and heterogeneous camera parameters. As a result, optimization-based SfM remains the standard solution for 3D reconstruction for large-scale unstructured image collections despite their high computational cost and sensitivity to initialization.

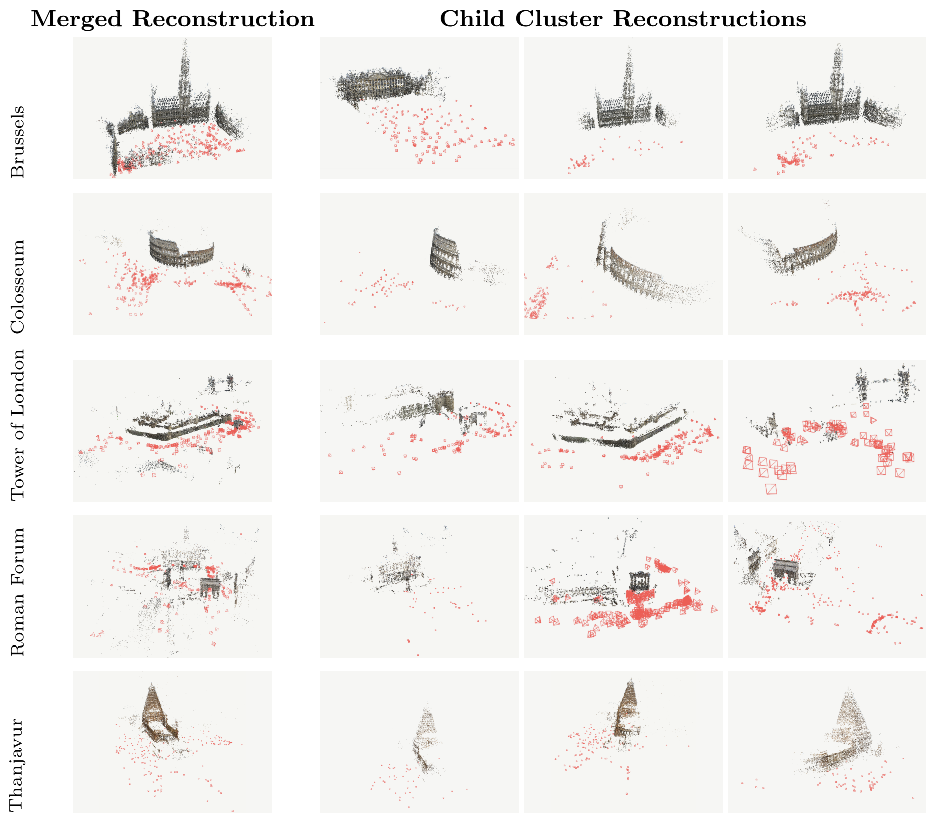

Fig. 2: Qualitative results. Final merged reconstructions (left) and individual cluster reconstructions (right) on Grand Place Brussels (234 images), Colosseum Exterior (499), Tower of London (1,576), Roman Forum (2,364), and Thanjavur (529). Clusters are reconstructed independently and joined by the bottom-up merge.

In this work, we propose a hierarchical reconstruction framework that enables modern multi-view 3D foundation models to operate on large-scale, unstructured image collections, as shown in Fig. 1. We employ nested dissection [16, 25] to recursively partition the verified view graph into balanced, overlapping clusters. The resulting separator cameras simultaneously define both the overlap required for hierarchical reconstruction and the separator variables of the underlying sparse bundle-adjustment system, aligning the reconstruction hierarchy with its computational structure. This allows each partition to be reconstructed independently while preserving sufficient geometric connectivity for global reconstruction. As a result, our divide-and-conquer formulation overcomes the scalability limitations of current foundation models, naturally supports parallel and distributed execution, and provides a principled framework for hierarchical reconstruction. Within this framework, we leverage a multi-view foundation model for fast camera initialization and combine it with classical geometric verification and hierarchical bundle adjustment to produce an accurate and globally consistent reconstruction. In summary, our contributions are as follows.

- We introduce a hierarchical SfM framework that enables modern multi-view 3D foundation models to reconstruct large-scale, unstructured image collections beyond their native memory limits.
- We formulate large-scale reconstruction as a divide-and-conquer problem by partitioning the verified view graph with Nested Dissection, enabling scalable and parallel reconstruction while preserving global geometric consistency.
- We bridge learned geometry and classical SfM by integrating foundation-model inference with geometric verification, hierarchical bundle adjustment, and cluster merging in a unified pipeline.
- We provide empirical insights into the interplay between learned priors and geometric optimization, and demonstrate competitive performance on the IMC PhotoTourism and 1DSfM benchmarks.

2 Related Work

2.1 Classical Structure from Motion

Classical Structure-from-Motion (SfM) pipelines estimate camera poses and 3D structure from image correspondences, typically followed by bundle adjustment (BA), which is the computational cornerstone of most SfM systems, to refine camera and point parameters jointly. Incremental SfM methods, such as COLMAP [32, 33], build a 3D model by sequentially registering images and triangulating new points. While robust, incremental approaches are prone to error accumulation and drift over long sequences [47]. In contrast, Global SfM methods solve for all camera poses simultaneously, offering greater efficiency and theoretical accuracy by avoiding drift [47]. Systems like GLOMAP have shown state-of-the-art results, outperforming incremental pipelines in both speed and accuracy on large-scale benchmarks [27].

The most recent trend is the application of deep learning to create end-to-end differentiable SfM pipelines. Classical systems consist of a series of hand-crafted, non-differentiable algorithms. In contrast, learned approaches like VGGSfM [37] replace this entire pipeline with a single neural network. This paradigm shifts the problem from online optimization to offline-trained inference, potentially sidestepping the runtime complexity of traditional BA.

Our method is motivated by the decomposition-based scaling literature discussed in Sec. 2.3, but differs in how local subproblems are initialized and merged, and leverages GTSAM [10] for factor-graph bundle adjustment.

2.2 Foundation Models

3D foundation models have ushered in a new era of scene reconstruction by leveraging large-scale pretraining and generalizable geometric reasoning. Recent models demonstrate impressive performance across diverse scenes and input conditions. DUSt3R [38] and MASt3R [22] estimate relative camera poses and dense point maps directly from image pairs. Building upon these pairwise formulations, foundation models extend to larger input batches through multi-view attention mechanisms, achieving more holistic scene understanding [19, 34, 35, 43, 46].

Despite these advances, existing models still struggle to scale to large or complex scenes due to GPU memory constraints, accumulated drift, and limited global consistency [5, 14, 24, 35]. Concurrent works have explored ways to alleviate these issues. Some focus on model efficiency, incorporating techniques such as quantization [14] or test-time adaptation [5], while others attempt to extend reconstruction coverage by feeding small portions of the scene sequentially to the model and merging the partial results [5,12,24]. However, scalability and flexibility remain open challenges [28]. Even with improved inference efficiency, memory limitations persist, and handling unstructured or unordered input remains underexplored. To address these challenges, we propose a divide-and-conquer strategy that partitions large-scale, unstructured image collections into manageable subsets, enabling efficient, scalable, and globally consistent reconstruction.

2.3 Scaling reconstruction: view-graph sparsification and partitioning

Exhaustive pairwise matching is quadratic in collection size and quickly prohibitive at internet scale. Two complementary reductions exist: sparsifying the *view graph* (which pairs are matched at all), and partitioning the *reconstruction problem itself* (which cameras are optimized together).

View-graph construction. Early systems selected candidate pairs by retrieval over visual vocabularies [26] or compact aggregated descriptors such as VLAD [17], forming a k-NN graph in descriptor space and pruning false candidates with RANSAC-verified two-view estimation - the retrieval+verification recipe popularized at scale by COLMAP [32]. Learned global descriptors [1, 4] improved robustness to viewpoint and illumination change; we adopt the localization-oriented MegaLoc [3], whose top-k proposals retain comparable accuracy while matching a fraction of the exhaustive pair budget.

Partitioned reconstruction. A complementary strategy for scaling SfM is to decompose large problems into smaller subproblems that can be solved independently and merged. HyperSfM [25] formalizes this idea through recursive hypergraph partitioning, producing a nested dissection hierarchy of constrained submaps separated by shared variables. Subsequent systems extend this divide-and-conquer view to global reconstruction: distributed motion averaging and consensus-based bundle adjustment split cameras across workers while enforcing agreement constraints [13, 30, 44, 45, 47], while GraphSfM [6] couples image clustering with graph-based merging for parallel large-scale reconstruction. Across these methods, the challenge remains the interface between subproblems: partitions must be small enough to enable efficient local computation, yet share enough cameras or covisible structure to make subsequent alignment and global refinement well constrained. Additionally, complementary work improves the efficiency of the underlying optimization through GPU-accelerated BA [31], numerically stable square-root formulations [11], and inverse expansion methods [40].

This divide-and-conquer approach has also been revisited by recent works in the context of feedforward reconstruction models. Our closest related work, GLUEMAP [28], partitions the view-graph into highly local clusters, with each cluster containing all the neighbors of a single image (*i.e.* number of clusters = number of images). MERG3R [7] partitions scenes into pseudo-sequences based on visual overlap. Although these methods are designed to operate on unordered collections, they are primarily evaluated on large sequential datasets with high visual overlap and similar lighting. In contrast, our graph-partitioning framework produces far fewer clusters than images and targets large unordered collections with varying capture times, lighting, camera intrinsics, and transient objects.

3 Method

We introduce a hierarchical Structure-from-Motion pipeline that leverages the strengths of feedforward geometry transformers. Given a large collection of unstructured images, we first create a view-graph, with edges between images that are possibly overlapping. This is the only part of our pipeline where two-view geometry and classical SfM techniques are used; these components are not used in the subsequent global reconstruction. We then partition the view-graph using nested dissection based ordering to create a hierarchical cluster tree.

Each cluster is initialized with VGGT-Omega, after which verified SIFT tracks are triangulated and refined through local bundle adjustment and Graduated Non-Convexity (GNC) based robust loss optimization. The resulting local reconstructions are merged hierarchically via Sim(3) alignment, followed by a global bundle adjustment with pose prior and focal length priors before obtaining the final globally consistent reconstruction.

The complete pipeline is summarized in Algorithm 1, and the individual components are described in detail in the following subsections, including image retrieval and geometric verification (Sec. 3.1), hierarchical view-graph partitioning (Sec. 3.2), foundation model inference and local refinement (Sec. 3.3), cluster merging and global refinement (Sec. 3.4), as shown in Fig. 3.

3.1 View Graph Construction

Given a collection of unordered images, we first identify candidate image pairs using the MegaLoc image retrieval system, which efficiently predicts overlapping

Algorithm 1 Large-Scale 3D Reconstruction Using Hierarchical Partitioning

Input: Unordered images $\mathcal{I} = \{I_i\}_{i=1}^{N}$
Output: Global cameras $\mathcal{C}$ and sparse 3D points $\mathcal{X}$

Retrieve candidate image pairs $\mathcal{E}_r$ using MegaLoc
Extract SIFT features and match retrieved pairs
for all $(i, j) \in \mathcal{E}_r$ **do**
    Estimate relative pose with PoseLib
    Refine pair using two-view bundle adjustment
    Keep edge (i, j) if geometric verification succeeds
end for
Build verified graph $\mathcal{G}_v = (\mathcal{V}, \mathcal{E}_v)$
Construct global verified SIFT tracks from $\mathcal{E}_v$
Partition $\mathcal{G}_v$ using Nested Dissection
for all clusters $\mathcal{G}_c$ **do**
    Initialize cameras with VGGT-Omega
    Triangulate verified SIFT tracks
    Run per-cluster BA with GNC
end for
for all levels of the cluster tree, from leaves to root **do**
    for all parent clusters with reconstructed children **do**
        Align and Merge child clusters into parent using shared cameras/tracks
        Run merge BA without GNC
    end for
end for
Re-triangulate global verified SIFT tracks against the merged cameras
Run global BA with all priors released
return final reconstruction $(\mathcal{C}, \mathcal{X})$

image pairs while avoiding exhaustive pairwise matching. Feature correspondences are then established using SIFT descriptors for each retrieved pair.

Each candidate pair is subsequently verified through geometric estimation. Specifically, relative camera poses are estimated using PoseLib with robust estimation, followed by a two-view bundle adjustment that jointly refines the relative pose and scene structure. Pairs whose optimized reconstruction fails geometric consistency checks are discarded, and the remaining image pairs form a verified view graph. This verification stage significantly suppresses incorrect retrievals and mismatches before global optimization.

3.2 Hierarchical View Graph Partitioning

Direct optimization over the complete verified graph becomes computationally expensive for large-scale scenes. Therefore, we partition the verified graph into a hierarchy of overlapping clusters, enabling parallel reconstruction on each smaller cluster followed by cluster merging.

We retain the largest connected component from the view graph and construct a symbolic factor graph whose factors correspond to verified view graph edges. We then apply nested dissection ordering and perform symbolic multi-

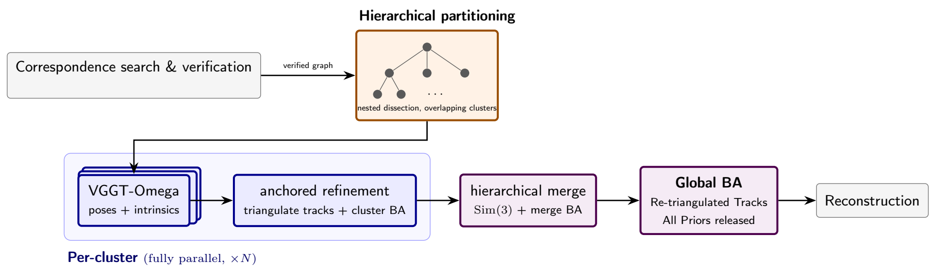

Fig. 3: Pipeline overview. MegaLoc retrieval prunes the candidate pair set before SIFT matching and two-view verification, producing a verified image graph, which is recursively partitioned into overlapping clusters by nested dissection. Each cluster is reconstructed by VGGT-Omega - poses and intrinsics taken directly from the model - with verified SIFT tracks triangulated against the predictions and refined by a pose-anchored bundle adjustment; clusters are processed fully in parallel, as are the bottom-up Sim(3) merges. A single unconstrained bundle adjustment over the re-triangulated global tracks produces the final reconstruction. Thick colored boxes mark our contributions; gray boxes are standard components.

frontal elimination, which produces a clique tree. Each clique contains a set of frontal cameras, which are assigned to the local cluster, and separator cameras, which are shared with ancestor cliques and provide overlap across the hierarchy.

Nested dissection ordering ensures that we obtain a balanced hierarchical partition of the verified graph while minimizing the number of edges cut between partitions. The resulting clusters contain strongly connected subsets of images that can be reconstructed independently while maintaining sufficient overlap for later merging and refinement.

3.3 Foundation Model Inference and Local Refinement

Each graph partition is reconstructed independently using the VGGT-Omega feedforward model [36]. VGGT-Omega simultaneously estimates camera poses and depth maps for all images within a cluster. Verified SIFT correspondences are used to form sparse 2D tracks, and their 3D landmarks are initialized from the predicted depth maps at the corresponding image locations, yielding an initial cluster reconstruction.

The reconstructed cameras and sparse points are subsequently refined with bundle adjustment. Let $\theta = \{\mathbf{T}_i, \mathbf{K}_i, \mathbf{X}_j\}$ denote the camera poses, intrinsics, and 3D points in a cluster. For an observation $\mathbf{u}_{ij}$ of point $\mathbf{X}_j$ in image i, the reprojection residual is

$$\mathbf{r}_{ij}(\theta) = \mathbf{u}_{ij} - \pi(\mathbf{T}_i, \mathbf{K}_i, \mathbf{X}_j), \tag{1}$$

where $\pi(\cdot)$ is the camera projection function.

During this stage, we impose priors on the estimated poses and intrinsics: 3D foundation models are highly accurate on low-radius visibility graphs [28], and our partitioning produces exactly such low-radius clusters. To improve robustness to remaining outlier observations, we optimize the reprojection error objective using Graduated Non-Convexity (GNC) [29,42] with a Truncated Least Squares (TLS) loss. GNC begins with a convex surrogate of the robust objective and iteratively increases the non-convexity so that the surrogate becomes closer to the true TLS loss, allowing outlier measurements to be downweighted and suppressing their influence during optimization. This produces an accurate and robust local reconstruction for each cluster.

Under GNC, we optimize the following objective as an iterative weighted least-squares problem.

$$\min_{\theta} \sum_{(i,j)\in\mathcal{O}} w_{ij}^{(k)} \left\| \mathbf{r}_{ij}(\theta) \right\|_{\Sigma}^{2} \tag{2}$$

where $\mathcal{O}$ is the set of feature observations.

For GNC-TLS, the weights are updated based on the current squared residual $s_{ij}^{(k)}$ as

$$w_{ij}^{(k)} = \begin{cases} 1, & s_{ij}^{(k)} \leq \frac{\mu}{\mu+1} c^2, \\ 0, & s_{ij}^{(k)} \geq \frac{\mu+1}{\mu} c^2, \\ \sqrt{\frac{\mu(\mu+1)c^2}{s_{ij}^{(k)}}} - \mu, & \text{otherwise}, \end{cases} \tag{3}$$

where c is the truncation threshold and μ is the control parameter that is increased across iterations. As μ grows, the surrogate objective approaches the target TLS loss, progressively downweighting high-residual observations.

3.4 Cluster Merging and Global Refinement

The independently reconstructed clusters are then merged into a unified global reconstruction. For every parent-child pair in the tree, we estimate a similarity transform Sim(3) using common cameras and overlapping tracks between the local reconstructions. This registers all the local reconstructions into the coordinate system of their parent.

After the initial alignment, duplicate tracks are merged, and global bundle adjustment jointly refines all camera poses, intrinsics, and 3D points. Similarly to individual cluster optimization, this stage incorporates pose and focal priors derived from the local reconstructions to preserve their geometric consistency while optimizing the global model. Since the majority of geometric outliers have already been removed by the two-view verification and local cluster optimization processes, we use standard nonlinear least-squares optimization without GNC at this stage.

Following GLOMAP [27], we then rebuild the sparse scene structure by triangulating globally verified 2D tracks with the merged camera poses, and perform a final bundle adjustment.

4 Experiments

We evaluate our method on a subset of scenes from the IMC PhotoTourism collections [8], and the large-scale 1DSfM internet collections [41]. These benchmarks cover datasets from under a hundred to nearly 3,000 images and encompass both images with EXIF-provided intrinsics and fully uncalibrated images. In addition to aggregate performance metrics, we present early observations into the strengths and limitations of our method.

Implementation. We retrieve candidate pairs with MegaLoc [3] (k=120 neighbors per image, similarity ≥ 0.15) and extract 8,192 SIFT features per image on GPU. Pairs are verified by 5-point estimation inside locally-optimized RANSAC [9, 21] as implemented in PoseLib [20] (2 px threshold), followed by a two-view bundle adjustment polish. We use the nested dissection implementation in METIS [18] for graph partitioning. Cluster merging selects up to 150 point correspondences per parent–child pair, preferring tracks with at least three observations and reprojection error below 1.5 px. The merge bundle adjustment applies pose priors (σ=0.1) and focal-length priors (σ=10 px) derived from the local reconstructions, and discards tracks whose reprojection error exceeds 3 px after optimization; the final bundle adjustment following global retriangulation releases all priors. All optimization uses GTSAM [10], with GNC-TLS [42] in the per-cluster BAs.

Metrics. On IMC PhotoTourism, we follow GLOMAP [27] and report area under the curve (AUC) of relative pose error over image pairs. The relative pose error is defined as the maximum of relative rotation error and relative translation (direction) angular error for an image-pair. On the 1DSfM scenes, where reference models are Bundler-era incremental SfM reconstructions with metric position ground truth, we follow the standard dataset protocol and report the number of cameras registered to the reference (N_c), along with the median and mean camera position errors ($\tilde{x}$ / $\bar{x}$, in meters) after robust Sim(3) alignment.

4.1 IMC PhotoTourism 2023 [8]

Despite the divide-and-conquer structure, our pipeline registers $\geq$95% of the images on every scene (100% on Sacré-Cœur and British Museum) and outperforms GLOMAP at AUC@5° and @10° on five out of the nine scenes, with margins up to +14.9 (Sagrada), +11.3 (Lincoln Memorial), and +10.2 (Sacré-Cœur) at 10°; on those three scenes it exceeds both GLOMAP and COLMAP at every threshold. These collections carry no EXIF metadata: intrinsics are taken from VGGT-Omega and refined by the anchored per-cluster adjustment of Sec. 3. The remaining gap is concentrated at the strictest threshold: optimization based global sfm retains an edge at AUC@3° on Pantheon, Piazza San Marco, and most visibly British Museum.

Table 1: Selected IMC 2023 Datasets (no EXIF). Relative-pose AUC (%) over all image pairs; images that fail to register count as failures. **Bold** marks the best method per scene and threshold. COLMAP and GLOMAP numbers are as reported in [27] (Table 5).

Scene	#**images**	#**reg. (ours)**	COLMAP [32]			GLOMAP [27]			**Ours**		
			AUC@3°	@5°	@10°	AUC@3°	@5°	@10°	AUC@3°	@5°	@10°
Piazza San Marco	68	65	58.0	71.0	83.7	**72.7**	**82.3**	**90.6**	59.6	70.5	80.1
St. Pauls Cathedral	142	141	72.7	78.9	85.0	**74.2**	**80.2**	85.8	64.5	75.6	**86.0**
British Museum	176	176	61.6	**72.7**	**83.9**	**62.0**	**72.7**	83.5	53.8	63.2	78.2
Lincoln Memorial Statue	214	213	67.2	71.5	75.3	68.4	72.5	76.0	**70.4**	**79.1**	**87.3**
Grand Place Brussels	234	231	68.7	76.6	84.2	71.8	78.5	84.6	**71.9**	**79.5**	**86.6**
Sacré-Cœur	281	281	78.5	80.9	82.8	78.8	81.1	83.0	**80.1**	**87.3**	**93.2**
Pantheon Exterior	320	319	**79.4**	**83.5**	87.2	77.0	81.8	86.2	73.6	82.6	**90.4**
Sagrada Família	397	377	54.3	58.9	62.9	53.8	58.7	62.9	**60.0**	**69.1**	**77.8**
Colosseum Exterior	499	494	**80.7**	**86.1**	90.1	80.5	85.8	**90.5**	66.4	76.1	85.4

Table 2: 1DSfM Internet collections. Cameras matched to the reference bundle (N_c; superscript: additional registered cameras absent from the reference), median and mean position error (m) after robust Sim(3) alignment, over all matched cameras. 1DSfM numbers are the robust-loss "1DSfM+BA" results of [41]. GLOMAP+CC is with camera-clustering enabled.

Scene	N/N_{ref}	**1DSfM** [41]			**GLOMAP** [27]			**GLOMAP+CC**			**Ours**		
		N_c	$\tilde{x}$	$\bar{x}$	N_c	$\tilde{x}$	$\bar{x}$	N_c	$\tilde{x}$	$\bar{x}$	N_c	$\tilde{x}$	$\bar{x}$
Alamo	2,915/761	529	0.30	2e7	$\mathbf{746}^{+465}$	0.36	39.8	637^{+100}	**0.29**	**3.8**	738^{+209}	0.44	4.6
Madrid Metropolis	1,344/431	291	**0.50**	70	$\mathbf{427}^{+328}$	1.40	9.9	275^{+37}	1.02	**6.3**	365^{+59}	1.93	9.8
Piazza del Popolo	2,251/455	308	2.10	7.0	$\mathbf{451}^{+683}$	0.38	5.1	395^{+427}	**0.36**	**1.1**	420^{+591}	0.47	1.9
Roman Forum	2,364/1,379	989	**0.20**	3.0	$\mathbf{1{,}366}^{+407}$	0.28	594	$1{,}124^{+190}$	0.23	**1.3**	$1{,}278^{+196}$	0.69	9.6
Tower of London	1,576/592	414	1.00	40	$\mathbf{586}^{+328}$	1.01	28.4	479^{+151}	**0.43**	**4.4**	584^{+278}	2.34	11.1

4.2 1DSfM: Large-Scale Internet Phototourism Collections [41]

We evaluate on five of the larger 1DSfM collections, whose multi-building layouts and repeated architecture make cross-scene connectivity sparse and ambiguous. Beyond the published 1DSfM results, we compare against GLOMAP [27] running on the *identical view graph as ours* - the image pairs proposed by our MegaLoc retrieval, exhaustively matched and geometrically verified with COLMAP - under both its default settings and with its camera-clustering postprocess enabled. This isolates the reconstruction backend: all systems consume the same images, features, correspondences, and verified two-view geometries.

Our method is able to balance both recall and mean position error. Against GLOMAP's default configuration, we match nearly the same number of reference cameras - within about 1% on Alamo and Tower of London, and 95% in aggregate over the five scenes - while our mean position error is lower on every scene, noticeably by up to 62× (Roman Forum). GLOMAP with camera-clustering enabled, bounds its errors by deleting 12–36% of the recoverable reference cameras (fragmenting Madrid into three components and Alamo into seven), we match more reference cameras on every scene and localize more within 10 m on four out of five scenes. (Tab. 3).

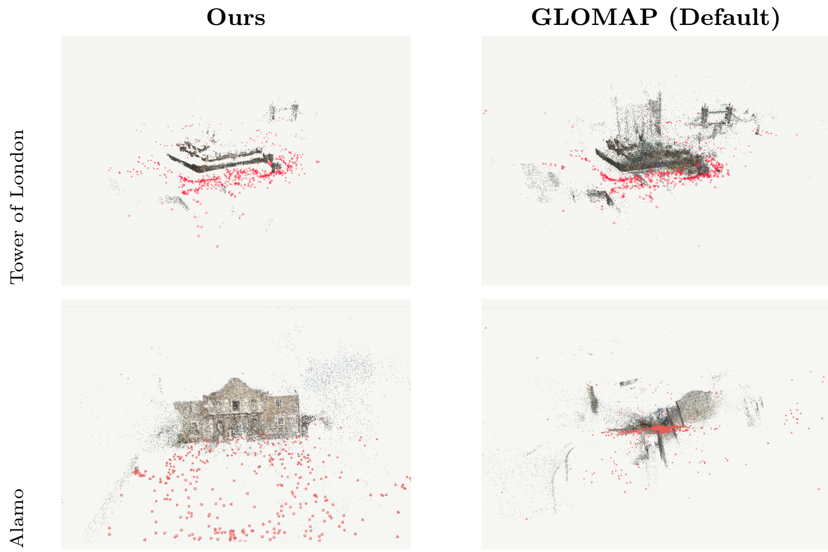

Fig. 4: Same view graph, two backends. Reconstructions from the identical MegaLoc-based view graph (GLOMAP under default settings). GLOMAP registers more cameras at better median accuracy (Tab. 2); the visible dispersion corresponds to its unbounded mean-error tail, removable via camera clustering only at the registration cost quantified in Tab. 3.

Table 3: Reference cameras localized within 10 m (% of N_{ref}). GLOMAP+CC and ours both suppress the failure tail

Scene	GLOMAP	GLOMAP+CC	Ours
Alamo	**93.2**	82.3	92.2
Madrid Metropolis	**86.8**	60.1	66.8
Piazza del Popolo	**96.5**	85.7	89.7
Roman Forum	**94.6**	79.8	78.8
Tower of London	**84.6**	74.7	79.7

4.3 Effect of Sparse View Graphs

Correspondence search is the dominant cost of classical SfM at scale. Our frontend verifies a retrieval-bounded pair list (k=120 nearest neighbors per image, similarity ≥ 0.15), making it $O(kN)$ rather than $O(N^2)$. Table 4 reports verified pairs against the $N(N-1)/2$ upper bound: on the IMC scenes the bound is barely active, while at 1DSfM scale we verify 2.8–5.8% of all pairs. We note that GLOMAP and COLMAP also employ retrieval-based candidate selection; the pair counts behind their published numbers are not reported, so Table 4 should be read as an absolute budget rather than a reduction relative to those baselines.

4.4 Where should a hybrid pipeline place its trust?

Scope. We make the following observations from our experiments. Observation 1 is measured on a single uncalibrated dataset (British Museum, 176 images); Ob-

Table 4: Verified pair budget with Megaloc Retrieval vs. exhaustive matching

Scene	#images	exhaustive	ours	%
Piazza San Marco	68	2,278	1,942	85%
British Museum	176	15,400	13,842	90%
Lincoln Memorial Statue	214	22,791	18,329	80%
Grand Place Brussels	234	27,261	14,585	54%
Sacré-Cœur	281	39,340	26,153	66%
Pantheon Exterior	320	51,040	31,016	61%
Sagrada Família	397	78,606	18,210	23%
Colosseum Exterior	499	124,251	51,003	41%
Madrid Metropolis	1,344	902,496	51,979	5.8%
Tower of London	1,576	1,241,100	53,734	4.3%
Piazza del Popolo	2,251	2,532,375	97,949	3.9%
Roman Forum	2,364	2,793,066	153,154	5.5%
Alamo	2,915	4,247,155	117,998	2.8%

servation 2 on a single 68-image scene, chosen because it is small enough to reconstruct in one forward pass. Both use the same feedforward backbone (VGGT-Omega) and partitioning scheme (our nested dissection). Efficiency claims in this paper are structural: the frontend ($O(kN)$ complexity), cluster-level reconstruction, and leaf-level merging are trivially parallelizable. Each observation informed a design decision in our pipeline.

Table 5: Calibration sources vs. the reference (176 cameras, no EXIF).

Focal source (British Museum)	median err.
Heuristic init. $1.2 \times \max(w, h)$	24.7%
ViewGraph Self-Calibration [15]	7.5%
VGGT-Omega predicted	1.1%
Refined against VGGT-Omega poses	1.1%
+ final free BA	**0.5%**

Observation 1: Feedforward models provide robust intrinsic parameter estimates. On the uncalibrated British Museum scene, we evaluate each focal-length estimate against the reference intrinsics (Table 5). Classical two-view self-calibration [15] reaches a median error of 7.5% in our low-parallax setting. In contrast, the focal lengths directly predicted by VGGT-Omega achieve a median error of 1.1%, measured over 2,133 predictions from 38 clusters. Predictions for the same camera across different clusters exhibit a median variation of only 2.3%, indicating that the model's calibration estimates are both accurate and consistent across contexts. Moreover, anchoring cluster-level bundle adjustment to the predicted poses and optimizing focal lengths using SIFT tracks independently recovers the same 1.1% error. A subsequent unconstrained bundle-adjustment stage further reduces the focal-length error to 0.5%.

Table 6: Hierarchy ablation (Piazza San Marco). AUC over pairs of the identical 65-camera subset retained by all variants. VGGT-Omega forward pass retains all 68 cameras at 40.7/59.0/76.8

Piazza San Marco (65/68 images)	@3°	@5°	@10°
VGGT-Omega, single pass	41.7	59.9	77.5
+ retri. + free BA	62.6	74.0	84.8
partitioned (≤30) + merged	62.1	75.3	86.8
+ retri. + free BA	**65.2**	**77.2**	**87.8**

Observation 2: Hierarchical partitioning remains beneficial even when it is not computationally necessary. Piazza San Marco contains only 68 images and can therefore be reconstructed in a single VGGT-Ω forward pass, allowing us to isolate the contribution of the hierarchical decomposition itself (Table 6). Retriangulation followed by a final unconstrained bundle adjustment substantially improves the one-shot prediction, increasing performance from 41.7 to 62.6 AUC@3°. Imposing a partition into clusters of at most 30 cameras initially produces weaker cluster-level reconstructions, with an average accuracy of 25.8 AUC@3°. Nevertheless, after merging and applying the same final refinement, the hierarchical reconstruction outperforms the one-shot alternative at every evaluation threshold. This result is consistent with the observation of [28] that feed-forward reconstruction accuracy degrades with increasing graph radius: smaller image sets support more accurate local predictions, while overlapping cluster merges and a subsequent unconstrained global optimization reconcile these local estimates into a more accurate global solution.

5 Conclusion

In this paper, we present a hierarchical SfM framework that integrates multi-view transformer-based 3D foundation models with classical geometric optimization for large-scale reconstruction. By combining verified image correspondences, graph partitioning, VGGT-Omega initialization, and hierarchical bundle adjustment, our approach leverages the efficiency of foundation models while retaining the accuracy and robustness of optimization-based SfM. The proposed divide-and-conquer framework naturally supports parallel reconstruction, enabling scalable reconstruction of large, unstructured image collections.

Nevertheless, our method still relies on classical SfM components, including traditional feature matching, geometric verification, and bundle adjustment, which remain computational bottlenecks. Furthermore, reconstruction quality depends on the underlying foundation model and the quality of graph partitioning. Future work can explore tighter integration between foundation models and geometric optimization, utilizing the inherent noise suppression properties of foundation models to operate on un-verified image similarity graphs, and more robust partitioning strategies.

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